Q.1 Compare multiprogramming and multitasking operating system.

Multiprocessing vs Multitasking

Feature	Multiprocessing	Multitasking
CPU	Uses two or more CPUs/processors	Uses only one CPU
Simultaneity	Tasks literally run at the same time	Tasks take turns rapidly (switching)
Speed	Faster, handles more jobs at once	Slower than multiprocess
Efficiency	More efficient and powerful	Moderate efficiency
Cost	More expensive (needs more hardware)	Cheaper (wo on single CP
Example	Servers, supercomputers	Personal computers, laptops

Q.2 Explain critical section in detail.

A "critical section" is a part of a program where shared resources (like data or files) are accessed. Only one process can enter at a time to prevent errors or data corruption. Operating systems use locks or semaphores to control access and make sure processes don’t interfere with each other.

A critical section should satisfy these conditions:

Mutual Exclusion: Only one process is allowed in the section at a time.

Progress: If no process is in the section, others can enter.

Bounded Waiting: Every process will eventually get its turn and not wait forever.

Q.3 Types of operating systems.

include:

1. Batch Operating System

2. Time-Sharing Operating System

3. Distributed Operating System

4. Real-Time Operating System

5. Network Operating System

6. Embedded Operating System

Q.4 Identify (Discuss) Functions of operating system.

Manages hardware resources:

Controls things like the CPU, memory, and devices so programs can use them easily.

Runs and schedules programs:

Starts and stops programs and decides which program should run next.

Organizes files and directories:

Helps store, find, and protect files and folders on your computer.

Handles input/output:

Helps the computer work with keyboards, screens, printers, and other devices.

Provides security and user access control:

Keeps your data safe and makes sure only approved users can access certain information.

Offers a user interface:

Gives users ways to interact with the computer, such as icons, menus, or ttypin commands.

Q.5 Define Operating system and explain components of computer system.

Operating System Definition:

An operating system (OS) is software that manages computer hardware and allows users and programs to use the computer easily. It acts as a link between users, applications, and the hardware .

Components of a Computer System:

Hardware: The physical parts like CPU, memory (RAM), hard disk, monitor, keyboard, and mouse.

Operating System: System software that controls hardware and runs applications.

Application Programs: Software like browsers, games, and word processors that do specific jobs.

Users: People who use the computer

Q.6 List strategies for dealing with deadlocks and explain any two of them.

Strategies for dealing with deadlocks:

- Deadlock Prevention: Blocks at least one condition needed for deadlock, like making processes request all resources at once.

- Deadlock Avoidance:Checks if resource allocation is safe before allowing it, using algorithms like the Banker's Algorithm.

- Deadlock Detection and Recovery: Lets deadlock happen, then finds and fixes it.

- Deadlock Ignorance: Simply ignores deadlocks, hoping they won't occur.

Q.7 Explain process life cycle with diagram.

The process life cycle in an operating system describes the steps a process goes through from creation to termination.

New - Ready - Running - Waiting - (back to Ready) - Terminated

The main stages are:

New: Process is created.

Ready: Process is waiting for CPU.

Running: Process is being executed by CPU.

Waiting/Blocked: Process is waiting for some event or resource (like input/output).

Terminated/Exit: Process has finished its task and is removed from memory.

Q.8 Calculate Average turnaround time and average waiting time using First come first serve (FCFS) algorithm. Draw Gantt.

(NO ANSWER)

Q.9 Define Memory management and write its techniques.

Memory management is the operating system's task of handling and organizing computer memory so programs can run smoothly and efficiently.

Techniques include:

Paging: Divides memory into small, fixed-size blocks for easy access and efficient use.

Segmentation: Splits memory into sections based on program structure, like functions or data.

Virtual Memory: Uses both RAM and hard disk to pretend there is more memory, allowing large programs to run.